

Executive Performance & Revenue Intelligence Report

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From: Business Intelligence & Revenue Operations Team

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Subject: NovaTech Unified Revenue Architecture, Operational Insights, and Generative BI

Executive Summary

NovaTech's revenue engine was evaluated across three core operational silos: marketing lead acquisition, sales pipeline conversion, and customer post-sale support. Across **2,240 leads**, **\$12.36M in marketing spend** yielded **\$1.13M in directly attributed revenue**. The sales pipeline processed **499 closed deals**, generating **\$0.71M in closed won revenue** at a **63.13% win rate** with an average sales velocity of **67 days**. On the customer health side, **3,000 support tickets** averaged **58.2 hours** to resolve, with enterprise accounts logging **757 tickets** characterized by a **25.5% negative sentiment rate** and critical product incidents reaching resolution times over **102 hours**.

This report summarizes our data consolidation strategy, the design rationale behind the three-sheet interactive QuickSight dashboard, the impact of Generative BI (QuickSight Q) semantic modeling, and strategic recommendations to eliminate operational leakage and safeguard annual recurring revenue (ARR).

1. Data Strategy & Architecture

NovaTech previously operated across isolated source systems: marketing campaign trackers, CRM pipeline stages, and customer service desks. This fragmentation prevented cross-functional attribution, obscuring whether high-acquisition channels actually generated retained customer value.

- **Anchor Schema Design:** The data architecture established `novatech_crm_deals` as the transactional backbone. Using left outer joins, `novatech_marketing_campaigns` and `novatech_support_tickets` were unified on the primary entity key `account_id`.
- **Grain Integrity & SPICE Ingestion:** Joining individual ticket events (3,000 records) to CRM deals (499 records) introduced fan-out risks where deal amounts could inadvertently multiply during aggregation. By staging calculations in SPICE and maintaining dedicated single-domain views alongside the NovaTech Unified Dataset, we preserved metric integrity across financial and operational boundaries.
- **Calculated Business Metrics:** Core key performance indicators—such as marketing return on investment, sales velocity (`days_to_close`), binary win conversion (`is_won_deal`), and ticket resolution durations—were standardized directly inside the data model to prevent formula discrepancies between BI visuals and natural language interfaces.

2. Dashboard Design Rationale

The executive dashboard is structured into three dedicated functional views designed to guide leadership from top-of-funnel acquisition down through customer retention:

- **Sheet 1: Marketing Funnel:**
 - *Purpose:* Evaluate acquisition efficiency and spend-to-revenue ratios across campaigns and channels.
 - *Design:* High-level KPI summary cards across the top anchor gross spend (\$12.36M) and attributed returns (\$1.13M). Funnel progression and clustered bar visuals highlight lead drop-off and channel response divergence. Interactive controls allow immediate filtering by campaign channel and customer segment.
- **Sheet 2: Sales Pipeline:**
 - *Purpose:* Assess sales velocity, conversion probabilities, and primary pipeline loss factors.
 - *Design:* Tracks realized deal value (\$0.71M) and conversion rates (63.13%). One-click visual actions enable executives to click the "Won" deal stage or a specific industry bar (e.g., Technology or Healthcare) to cross-filter the entire sheet in real time.
- **Sheet 3: Customer Health:**
 - *Purpose:* Identify post-sale account friction, technical debt bottlenecks, and renewal vulnerabilities.
 - *Design:* Displays SLA performance across product categories and sentiment distributions across subscription tiers. A cross-dataset visual links CRM contract values with support ticket volumes. A

cross-sheet navigation action allows reviewers to click an at-risk enterprise account and jump directly to its pipeline history on the Sales sheet.

3. Generative BI: QuickSight Q Topic Configuration & Accuracy

To provide conversational self-service analytics for executives, we benchmarked Amazon QuickSight Q before and after semantic Topic optimization.

Baseline Performance (Out-of-the-Box Q)

Prior to semantic modeling, Q struggled with domain-specific terminology:

- Queries regarding **"win rate"** failed or generated raw deal counts because the system looked for an exact column match rather than averaging the binary win indicator.
- Inquiries regarding **"revenue"** prompted ambiguous disambiguation prompts, unable to distinguish between marketing attributed revenue and closed deal values.
- Technical database keys (account_id, opportunity_id, ticket_id) added noise to responses, occasionally causing Q to attempt arithmetic sums on string identifiers.

Topic Optimization Strategy

We engineered the NovaTech Business & Revenue Intelligence Topic with four targeted adjustments:

1. **Excluding Noisy Keys:** System IDs were toggled off, forcing Q to focus exclusively on business dimensions and operational measures.
2. **Business Synonyms:** Enriched core measures with executive vocabulary (e.g., mapping deal_value to *Annual Contract Value (ACV)*, *Annual Recurring Revenue (ARR)*, *pipeline value*, and *bookings*; mapping resolution_time_hours to *Service-Level Agreement (SLA)*, *Mean Time to Resolve (MTTR)*, and *ticket duration*).
3. **Semantic Instruction & Metric Mapping:** Mapped Win Rate directly to the average of is_won_deal formatted as a percentage.
4. **Natural Language Custom Rules:** Explicit instructions guided Q to prioritize closed won revenue for sales questions and attributed returns for marketing queries.

Measured Impact on Accuracy

Re-testing the baseline prompts demonstrated immediate structural improvements:

- Conversational prompts such as *"Show me win rate by industry"* returned an accurate visual ranking matching our **63.13%** benchmark.
- Revenue channel queries seamlessly grouped revenue_attributed without manual disambiguation.
- Complex filters, such as identifying campaigns where costs exceeded returns, were correctly computed using the underlying comparative logic.

4. Generative AI vs. Dashboard Comparative Analysis

To validate QuickSight Q's readiness for executive self-service, we cross-checked natural language query responses against the dashboard ground truth:

- *Where AI agreed with the Dashboard?* QuickSight Q achieved full alignment on single-domain aggregations and rankings, and dimensional segmentations. In marketing, Q correctly identified Direct Mail as the highest converting channel (53.0% conversion rate) while noting Email lagged at 8.7%. In sales pipeline analysis, Q verified that Enterprise accounts yield the highest average deal value. In customer support, Q accurately parsed operational duration splits across ticket priorities, reporting an average resolution time of 61.3 hours for critical tickets versus 58.4 hours for low-priority tickets, precisely matching the dashboard's 2.9-hour variance.
- *Where AI disagreed or struggled?* QuickSight Q encountered boundaries when processing multi-metric queries across different entity grains. When asked to list the top 10 accounts by support ticket volume alongside their total CRM deal revenue, Q could not return an answer.

- Grain and Visual Scope Limitations: The Customer Health sheet pre-aggregates operational data at higher categorical grains (product area, priority, customer tier, and industry) rather than at the individual account level (account_id), leaving no account-grain visual for direct cross-checking.
- Cross-Entity Multi-Metric Joins: querying across distinct entity boundaries (CRM deal revenue combined with support ticket counts at the account level) requires complex multi-table joins and grain reconciliation. Without a pre-built account summary table on the sheet, Q could not resolve the dual-metric ranking, documenting this as an intentional and valid finding.
- *When to use Q vs. the published dashboard?* Use QuickSight Q / Quick Chat for ad-hoc and comparative inquiries without building dedicated charts, unplanned dimensional slicing, and self-service exploration for non-technical users asking plain-English questions. Use the published dashboard for standardized executive monitoring, contextual decision-making, and guided cross-sheet workflows using interactive filters and navigation actions.

5. Key Findings & Strategic Recommendations

Area	Quantified Finding	Business Impact	Strategic Action
Marketing Efficiency	\$12.36M spend yielded \$1.13M revenue. Direct Mail converts at 53.0% vs. Email at 8.7%.	Low-touch email blasts dilute budget, high-touch targeted channels convert.	Reallocate \$500K from email to direct mail.
Sales Conversion	63.13% overall win rate. Pricing & Feature Gaps account for >60% of 184 lost deals.	Enterprise deals stall in late stages despite delivering highest <i>Annual Contract Value (ACV)</i> .	Deploy deal-desk pricing waivers & battle cards.
Customer Retention	3,000 tickets averaged 58.2h <i>Mean Time to Resolve (MTTR)</i> , Analytics Dashboard criticals exceeded 102h <i>Service-Level Agreement (SLA)</i> .	Product instability in key areas threatens enterprise ARR renewals.	Build Tier-3 escalation pod, cap <i>MTTR</i> at 24h.

Recommendation 1: Rebalance Marketing Acquisition Budgets

- **Observation:** Broad digital awareness campaigns underperform targeted outreach. The *NovaEdge Awareness* campaign incurred \$1.52M in spend while generating only \$34.3K in attributed pipeline. Conversely, *Direct Mail* and *Partner Referrals* delivered response rates of 53.0% and 34.3%.
- **Prescription:** Reallocate \$500,000 from underperforming broad email and awareness budgets directly into Account-Based Marketing (ABM), direct mail gifting, and co-marketing partner programs to drive higher-quality enterprise pipeline.

Recommendation 2: Institute an Enterprise Deal-Desk & Battle Card Library

- **Observation:** Sales cycle velocity averages 67 days, but enterprise accounts face significant friction. Post-mortem review of the 184 lost deals reveals *Pricing* objections and *Feature Gaps* as the predominant drivers of stalled pipeline.
- **Prescription:** Formalize an executive deal-desk workflow empowering sales leaders to approve structured multi-year pricing discounts and flexible payment terms. Concurrently, deliver competitive battle cards to equip account executives against rival feature comparisons.

Recommendation 3: Deploy an Enterprise Technical Escalation Pod

- **Observation:** Enterprise clients generated 757 support requests, with more than one out of four tickets expressing negative customer sentiment (25.5%). Critical system incidents in the *Analytics Dashboard* averaged over 102 hours to resolve—nearly double the organizational average of 58.2 hours.
- **Prescription:** Form a dedicated Tier-3 rapid engineering escalation pod tasked specifically with resolving enterprise outages. Enforce a strict 24-hour *MTTR SLA* for critical dashboard and authentication issues to protect recurring contracts ahead of annual renewal conversations.